

DiagAccU

Interface Documentation

Version 2.0.1

DOI: [10.5281/zenodo.22879016](https://doi.org/10.5281/zenodo.22879016)

Rallou A. Chatzimichail, MEng, MSc, PhD^a, Theodora Chatzimichail, MRCS^b, Aristides T. Hatjimihail, MD, PhD^c

^arc@hcsl.com, ^btc@hcsl.com, ^cath@hcsl.com

Hellenic Complex Systems Laboratory, Kostis Palamas 21, Drama 66131, Greece

Table of Contents

1. Introduction
2. Interface Controls
3. Input Variables and Parameters
4. Input Models and Assumptions
5. Notation
6. Citation
7. License

1. Introduction

This guide describes the inputs and controls available in the distributed, compiled DiagAccU 2.0.1 interface. Available inputs depend on the selected plot or table.

The test orientation is $Y \geq t$, where t is a specified fixed decision value with no assigned measurement uncertainty. Let $Y = X + \Delta$, where Δ is a common measurement bias relative to the scale of the fitted population distributions. Classification is equivalently $X \geq \tau$, with the diagnostic threshold expressed on the reference measurement scale, $\tau = t - \Delta$, nominally $\tau = t$. The population samples, prevalence, and common measurement bias are assumed to be mutually independent, with covariance retained between the estimated mean and standard deviation within each population. The constant coefficient b_0 , proportional coefficient b_1 , and selected model specify the standard uncertainty of this common shift.

2. Interface Controls

2.1. Tabbed Navigation

Choose a primary tab and then the required submodule. Each terminal view has its own controls and saved defaults; changing a control in one view does not synchronize every other view.

2.2. Numerical Settings: Sliders

The program offers controls for numerical settings, which can be adjusted through sliders. Press the 'alt' or 'opt' key while dragging the mouse for more precise control. For even finer adjustments, also press the 'shift' or 'ctrl' keys.

2.3. Non-Numerical Settings

Buttons and drop-down menus select the measure, distribution family, measurement-uncertainty model, and other settings. Available controls depend on the selected view.

2.4. Display Controls

2.4.1. Plots

Extended and limited settings principally change the range of the horizontal axis. Threshold plots span the population means plus or minus three or one standard deviations, respectively, with a nonnegative rounded lower bound. Prevalence profiles

use 0.001–0.999 or 0.01–0.2. Profiles of b_0 use 0–0.5 times s_H or 0–0.15 times s_H ; profiles of b_1 use 0–0.1 or 0–0.05. The plotted sample-size range is generated automatically. These settings do not specify the range over which the measure is optimized.

2.4.2. Tables

Table views provide a decimal-display control from 1 to 10, with a default of 3. This changes displayed precision, not the precision of the underlying calculation.

3. Input Variables and Parameters

The following ranges describe interface controls. Diseased and nondiseased populations are denoted D and H, and prevalence is v . Arithmetic means and Bessel-corrected standard deviations are supplied for all population families.

t : $\max(0.01, \min(m_H - 6s_H, m_D - 6s_D)) - \max(m_H + 6s_H, m_D + 6s_D)$

n_D : 2–10,000

m_D : 0.1–10,000

s_D : 0.01–1,000

n_H : 2–10,000

m_H : 0.1–10,000

s_H : 0.01–1,000

v : 0.001–0.999 in the views that expose prevalence; other views derive $v = \frac{n_D}{n_D + n_H}$ from the sample counts.

n_U : 20–10,000; interval views use this control to assign $v_{\text{meas}} = n_U - 1$ to the common measurement-uncertainty source.

b_0 : $0 \leq b_0 \leq s_H$

b_1 : 0–0.1

$1 - \alpha$: 0.900–0.999

n : 20–20,000; in the sample-size profile, the total sample size is varied automatically while prevalence is held fixed at the selected value.

The fixed cutoff, both means, both standard deviations, and b_0 must use the same measurement unit; b_1 is dimensionless. The common measurement bias, its standard uncertainty, and the transformed threshold use that same unit.

In uncertainty profiles parameterized by total sample size and prevalence, the program uses $n_D = vn$, $n_H = (1 - v)n$. Both resulting stratum sizes must exceed one for the core calculation to be valid. Thus some extreme prevalence–sample-size combinations within the slider ranges can produce unavailable results. Noninteger effective sample sizes can occur in calculations over the plotted range.

4. Input Models and Assumptions

A normal, lognormal, or gamma distribution is selected independently for each population. The supplied means and standard deviations are on the arithmetic scale.

The standard uncertainty of the common measurement bias is specified by a nonnegative linear model, $u_{\text{meas}} = b_0 + b_1 t$, or

the root-sum-of-squares form, $u_{\text{meas}} = \sqrt{b_0^2 + (b_1 t)^2}$. Here u_{meas} denotes $u_{\text{meas}}(t)$, evaluated at the fixed nominal cutoff.

The constant and proportional coefficients are supplied inputs. This interpretation requires empirical justification for a common component; total quality-control variance alone does not establish it. Independent analytical variation in individual observations must be represented in the measured population distributions or an explicit observation model without double counting.

The nominal diagnostic cutoff is supplied in fixed-threshold views and estimated automatically in optimized-threshold views. At each evaluation it is held fixed; measurement uncertainty is assigned to the common measurement bias, not to the chosen cutoff. Optimized intervals are conditional on the selected cutoff and exclude statistical uncertainty in its estimation and selection. The threshold slider and plot range do not specify the optimization limits. Existing software controls and labels

referring to threshold uncertainty retain their names for compatibility and denote the equivalent transformed-threshold representation of the measurement uncertainty component.

Interval views accept the nominal level $1 - \alpha$ and quality control (QC) sample count n_U . Changing n_U changes the degrees of freedom assigned to the measurement-uncertainty source and the coverage factor; it does not change the measurement standard uncertainty specified by the coefficients. A generic QC count does not by itself establish a common measurement uncertainty component or its degrees of freedom.

Prevalence uncertainty is included in the compiled interface. There is no separate control for excluding this contribution.

In the relative-standard-uncertainty profile against total sample size, the F_1 option is listed but does not produce a result.

5. Notation

5.1. Variables and Parameters

t : specified fixed nominal diagnostic cutoff; also used for the estimated optimal cutoff, held fixed in the conditional uncertainty calculation; Δ : common measurement bias; $\tau = t - \Delta$: transformed threshold

n : total sample size

v : prevalence

b_0 : constant contribution to the standard uncertainty of the common measurement bias

b_1 : proportional coefficient for the standard uncertainty of the common measurement bias

$1 - \alpha$: confidence level

5.2. Diagnostic Accuracy Measures

Se: sensitivity

Sp: specificity

ACC: overall diagnostic accuracy

PPV: positive predictive value

NPV: negative predictive value

DOR: diagnostic odds ratio

LR^+ : positive likelihood ratio

LR^- : negative likelihood ratio

J : Youden's index

BA: balanced accuracy

CZ: concordance probability

FMI: Fowlkes–Mallows index

κ : Cohen's kappa

PABAK: prevalence- and bias-adjusted kappa.

F_1 : F_1 score.

MCC: Matthews correlation coefficient

5.3. Uncertainty and Interval Outputs

u_c : combined standard uncertainty

u_s : total sampling standard uncertainty

u_m : measurement standard uncertainty of the selected measure from the common measurement bias

LCL: lower confidence limit

UCL: upper confidence limit

QC: quality control

6. Citation:

Chatzimichail RA, Chatzimichail T, Hatjimihail AT. DiagAccU: A unified framework for uncertainty estimation of diagnostic accuracy measures. Ver. 2.0.1. Hellenic Complex Systems Laboratory; 2026. DOI: <https://doi.org/10.5281/zenodo.22843673>

7. License

This document is licensed under the [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-nc-sa/4.0/).

First published: September 09, 2025.

Revised: September 21, 2026.